Experimental Science
The Scientific Revolution
1500 – 1700

Scientific Revolution Era

There was a belief in natural law and a conviction that mathematics was the language of Nature.

Strong rejection of authority, especially scholasticism

Commitment to observation and experimentation and a belief in a **mechanistic** cosmos.

There was the establishment of international communities of scientists; The Royal Society of London (1662) and the Academie des Sciences de Paris (1666).

Scientific Revolution Era

Information revolution and dissemination of it through the improvement of printing technology

It became increasing difficult for governments to control the dissemination of information, scientific or otherwise, as was the case in Galileo's Two New Sciences.

Science and Technology

- Technology seems to develop independently of science and run well ahead of our scientific understanding of the technological processes.
  - Steam technology developed well ahead of any understanding of thermodynamics.

Science and Technology

At other times, technological innovation clearly lays the foundation for scientific discovery. Invention of certain instrumentation - Galileo's telescopes: Galileo had no theory of optics to explain, or understand, how his telescope worked.

Even today, scientists spend a good deal of their time making their own instruments.

Telescope & Microscope

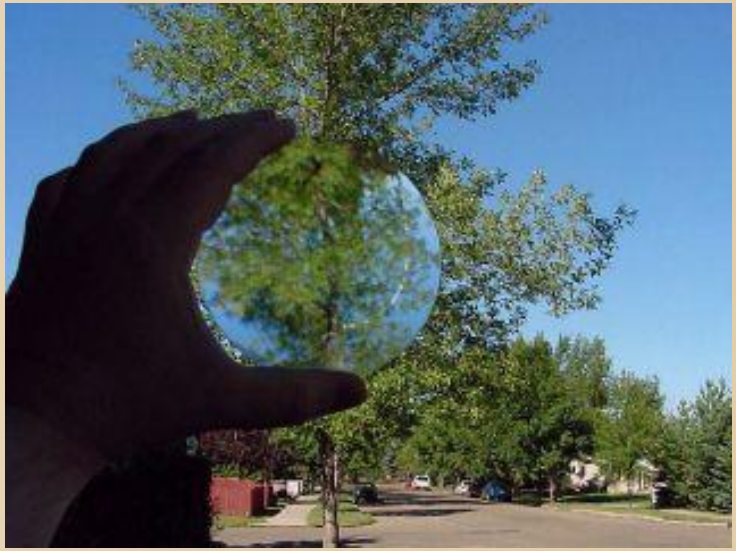
Galileo's Telescope (1609) gathers and focuses light, mainly from the visible part of the electromagnetic spectrum, to create a magnified image for direct view

The main problem with his telescopes was their very narrow field of view, typically about half the width of the Moon.

Optical Telescope

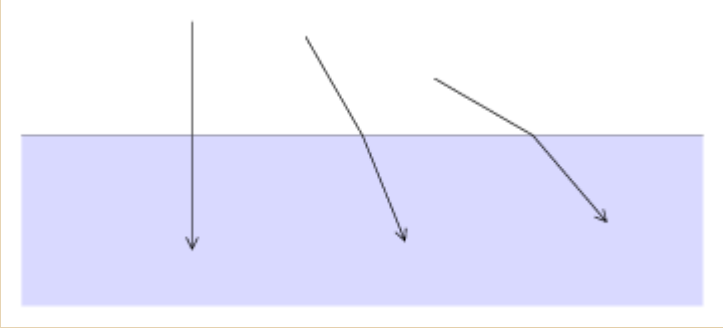
When you hold a magnifying glass a few inches away from a page of print, you can see the page enlarged.

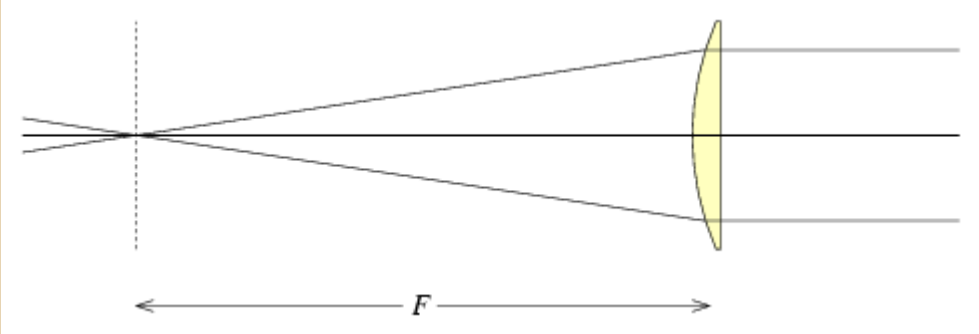
Also, when you look at a distant object through the same magnifying glass, you see an upside-down image of the object in front of the lens.



Optics

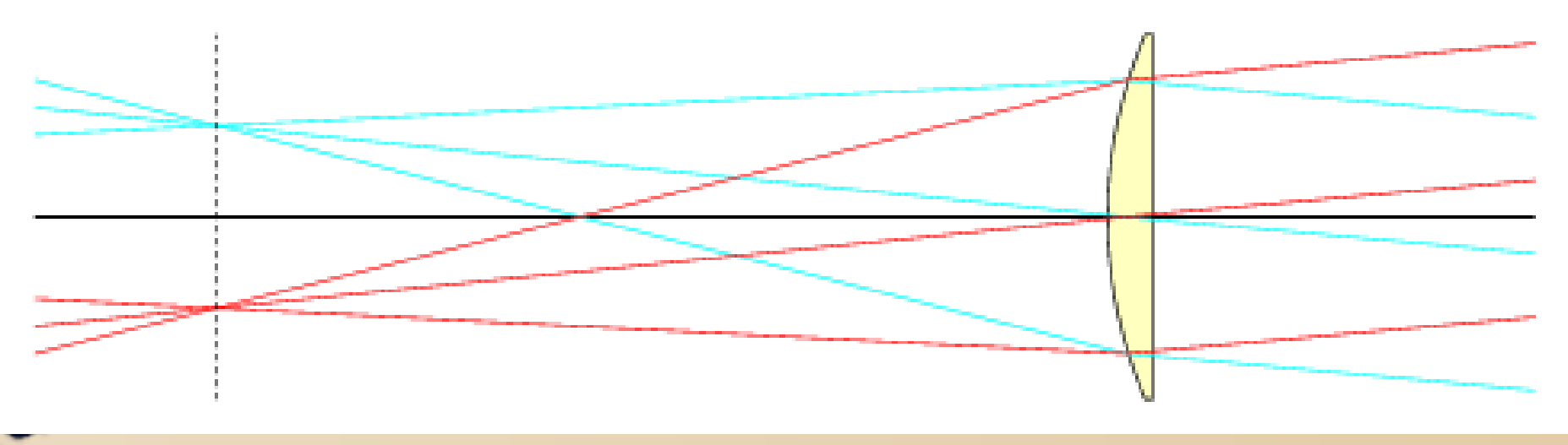
Light rays passing from air into water. If the ray strikes the surface at a 90° angle it does not bend (left). But if the angle is less than 90° the light does bend (middle); reducing the angle between the light and the surface increases the bend (right).





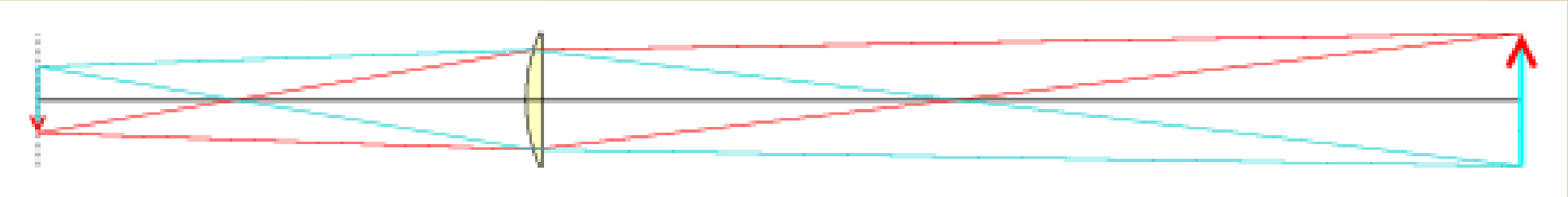
Object Lens

A simple lens forming an image. The red rays arrive with a downward slant, and come to a focus below the optical axis, while the blue rays arrive with an upward slant and come to a focus above the optical axis. The vertical dotted line at left represents the focal plane.



Object Lens

There's one slightly subtle consequence of this image-formation process: the image is upside-down! the focal plane.



Eye Lens

To make a telescope, you'll need to add another lens. This eyepiece lens magnifies the image formed by the large objective lens and directs the light to your eye

$$M = F \div f .$$

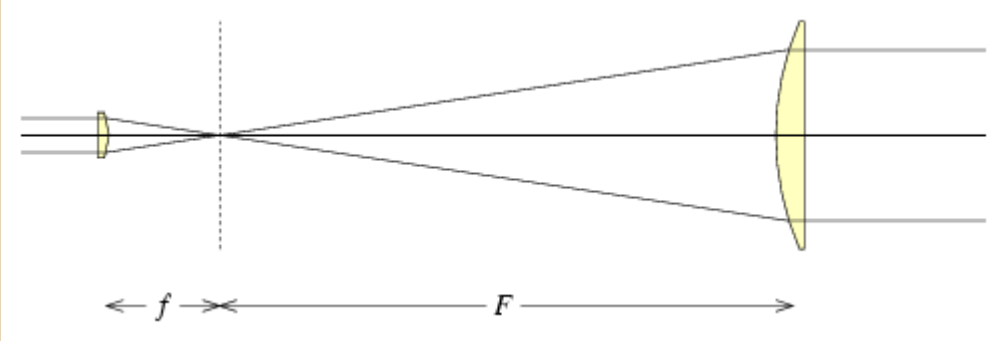
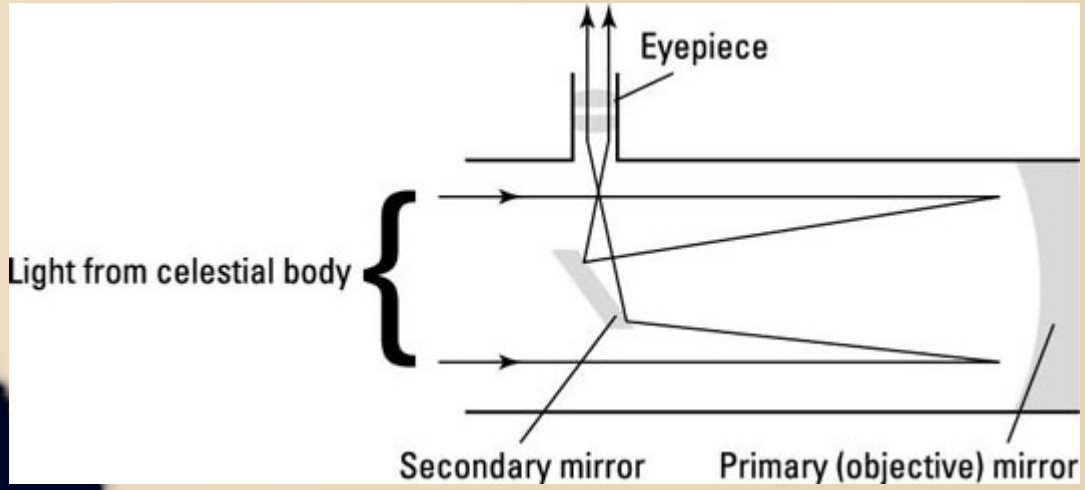
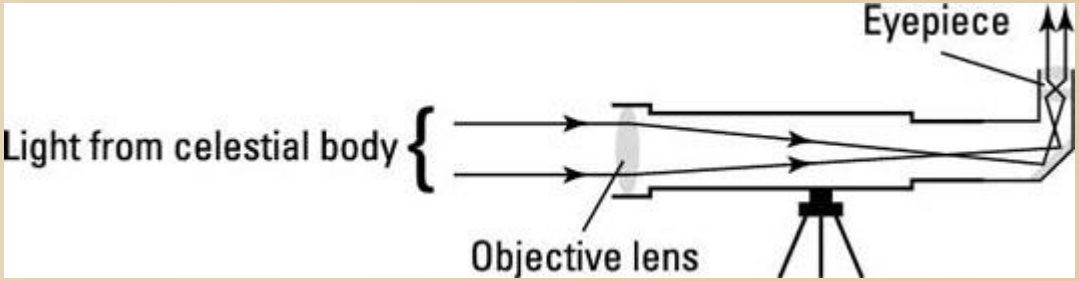


Diagram of a simple telescope. Parallel light rays enter from the left, pass through the objective lens, come to a focus at the focal plane, and exit through the eyepiece lens. The focal length of the objective is F, and the focal length of the eyepiece is f.

Reflecting vs Refracting

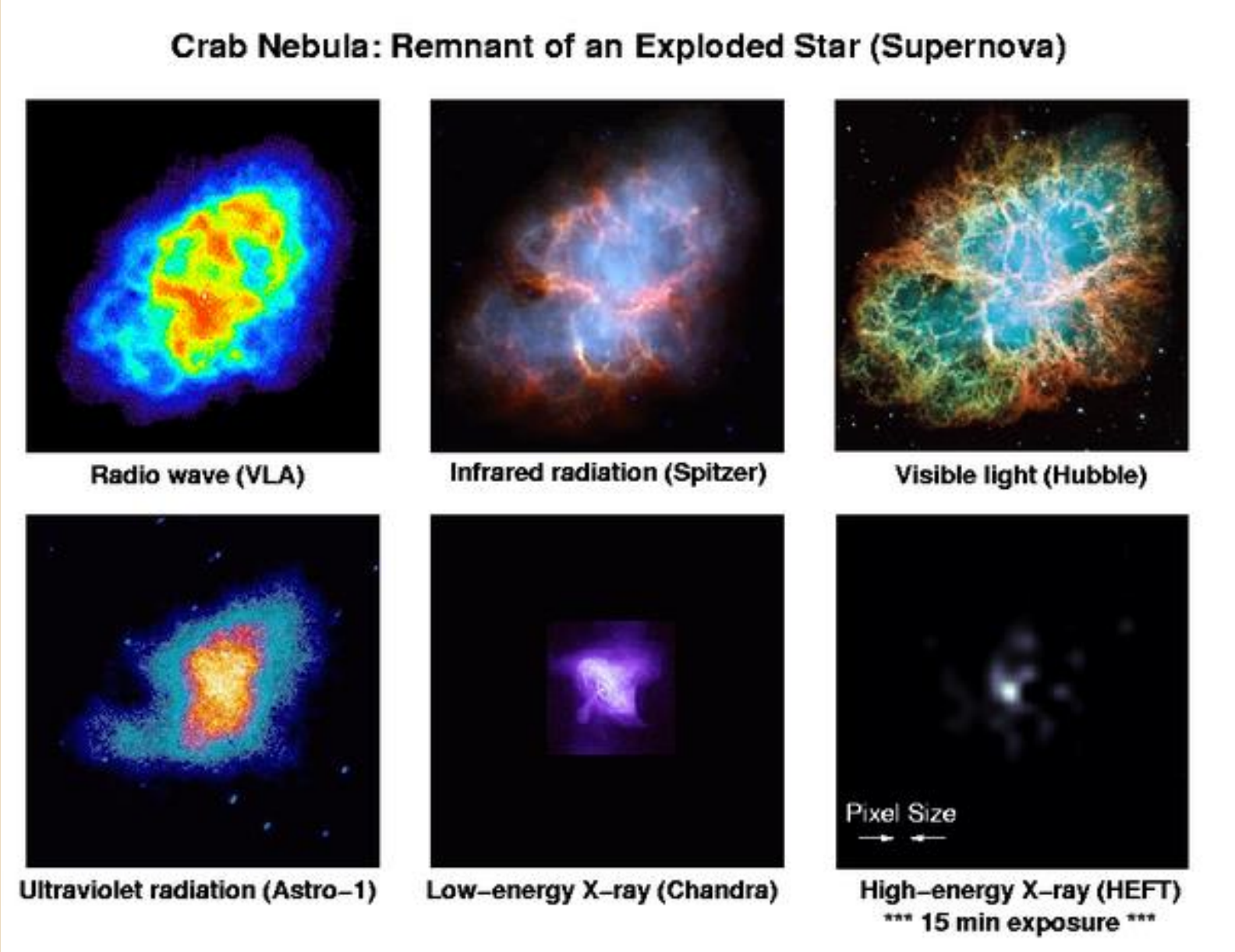
The light enters the tube and reflects off a curved mirror (called the primary mirror) at other end, before bouncing back up the tube to near the top where it reflects off a smaller mirror (called the secondary mirror), which bounces the light out of a hole in the side of the telescope where you attach your eyepiece.





The light enters a refracting telescope through the front lens, called the objective lens. It then travels down the length of the telescope to the eyepiece lenses, which is where the magnifying happens.

Kind of Telescopes

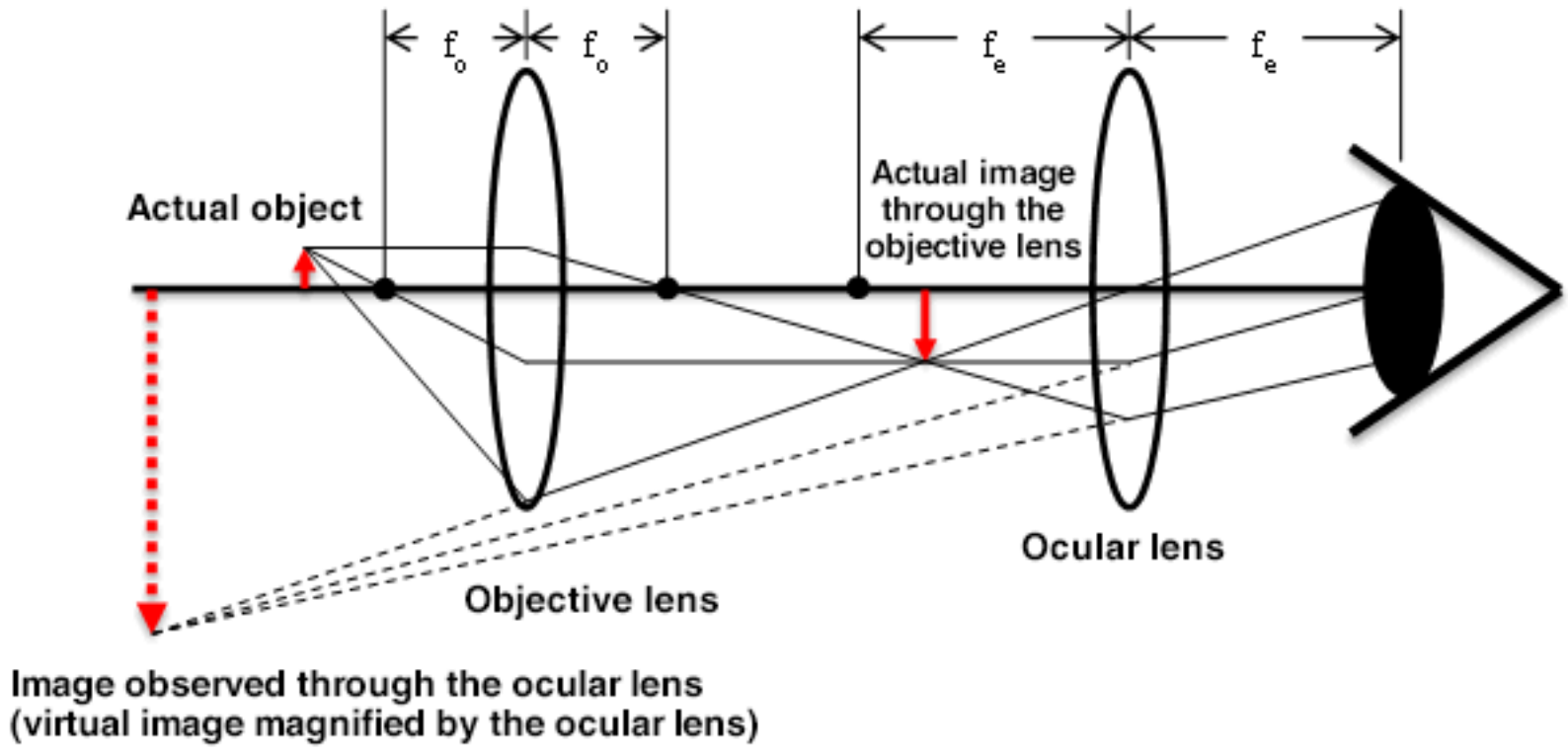


Microscope

A Dutchman, Antony van Leeuwenhoek (1632–1723), proved skilled in making microscopes. From a pail of lake water, he discovered scores of animalcules.

He had discovered protozoa, bacteria, and sperm. But it would not be until the 19 th century that Leeuwenhoek’s discoveries would lead to major advances in biology and medicine.

Microscope - Compound



Uses an objective lens close to the object being viewed to collect light which focuses a real image of the object inside the microscope. That image is then magnified by eye lenses which gives the viewer an enlarged inverted virtual image of the object

Todays Microscope

1. Eyepiece (ocular lens)
2. Objective turret, revolver, or revolving nose piece (to hold multiple objective lenses)
3. Objective lenses
4. Focus knobs (to move the stage)

   Coarse adjustment

5. Fine adjustment
6. Stage (to hold the specimen)
7. Light source (a light or a mirror)
8. Diaphragm and condenser
9. Mechanical stage

Kinds of Microscope

Electron Microscope

- Scanning Electron Microscope
- Transmission Electron Microscope

Atomic Force Microscopy - AFM

Electrostatic Force Microscopy – EFM

Watch Light and Dark – BBC Documentary

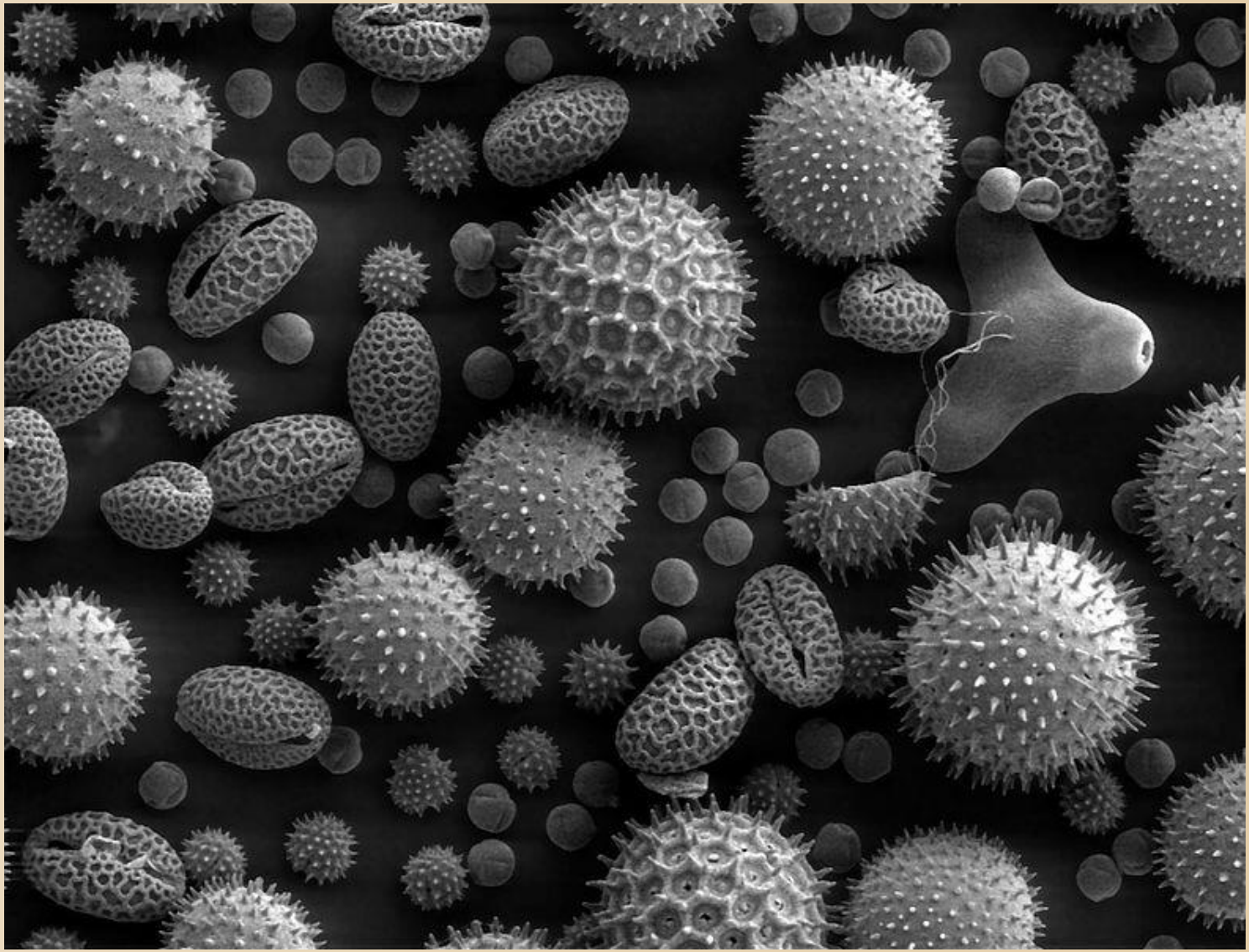
Scanning Electron Microscope (SEM)

Type of electron microscope that produces images of a sample by scanning the surface with a focused beam of electrons.

The electrons interact with atoms in the sample, producing various signals that contain information about the sample's surface topography and composition.

The electron beam is scanned in a raster scan pattern, and the beam's position is combined with the detected signal to produce an image. SEM can achieve resolution better than 1 nanometer.

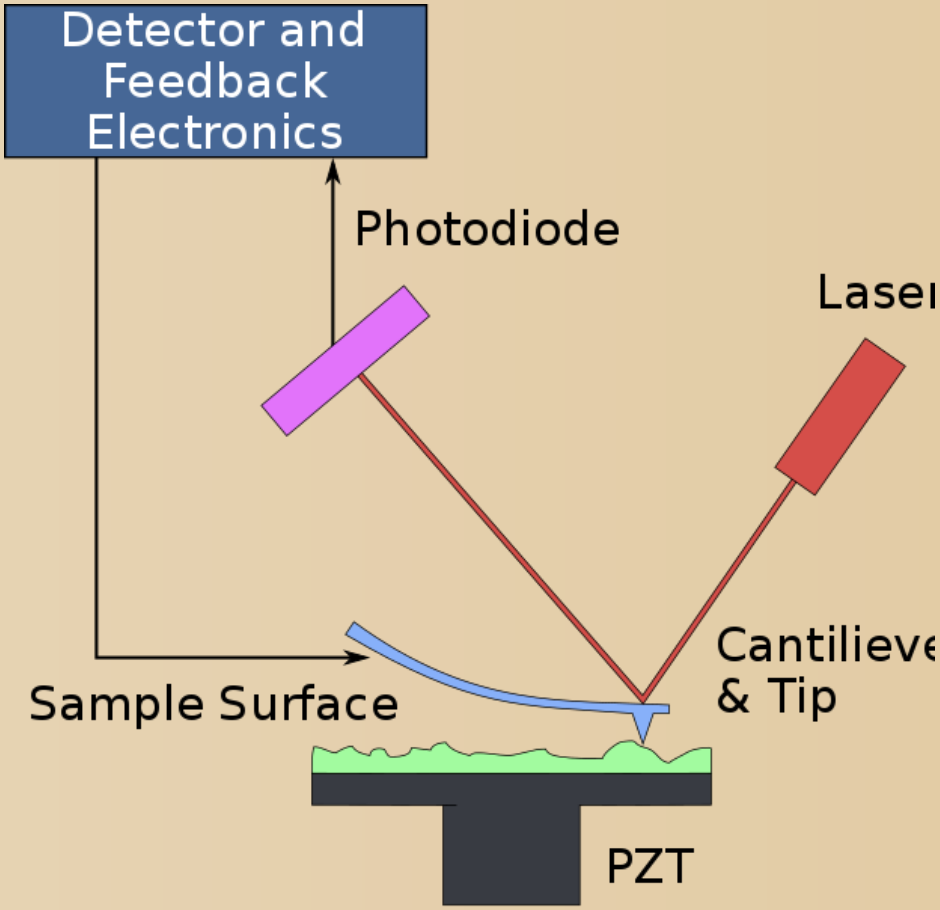
Scanning Electron Microscope (SEM)



Atomic Force Microscopy (AFM)

Type scanning probe microscopy (SPM), with resolution on the order of fractions of a nanometer.

The information is gathered by "feeling" or "touching" the surface with a mechanical probe. Piezoelectric elements that facilitate tiny but accurate and precise movements on (electronic) command enable very precise scanning.



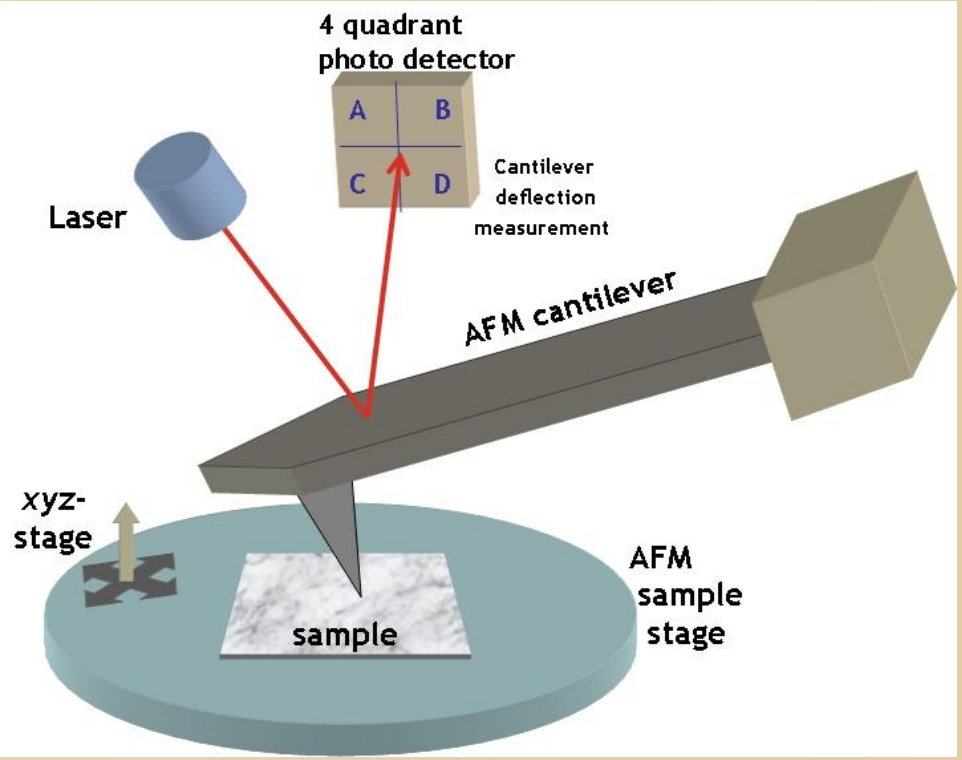
AFM Principle- Basic Training

Electrostatic Force Microscopy (EFM)

Type of dynamic non-contact atomic force microscopy where the electrostatic force is probed.

"Dynamic" here means that the cantilever is oscillating and does not contact the sample.

This force arises due to the attraction or repulsion of separated charges.



How AFM Works 3-2 Electrostatic Force Microscopy (EFM)

A Mechanistic World- 1600-1700

Rene Descartes (1596–1650) mathematician and founder of modern philosophy. He had to seek refuge in Holland which tolerated his unorthodox beliefs.

Until then mathematicians still worked rather cumbersomely with Euclidean geometry. Descartes' greatest contribution to mathematics was the invention of analytical geometry.

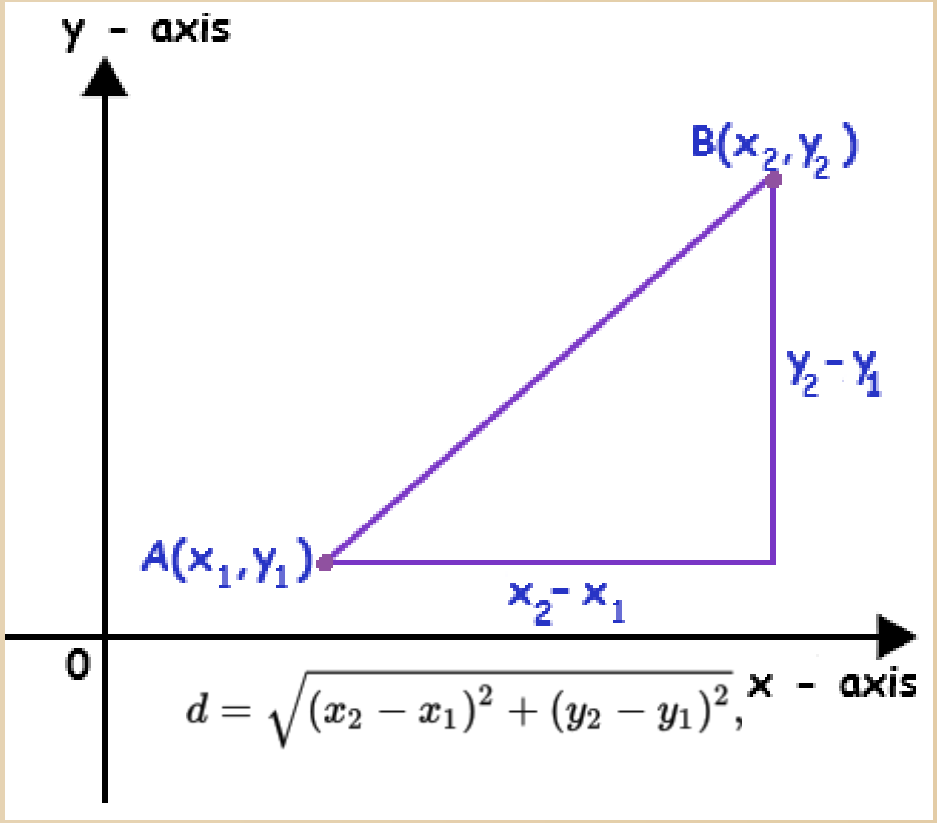
With Descartes' analytical geometry, one can test a new cannon on paper, with a high degree of accuracy, before firing the cannon in outside field tests.

Descartes and Location of a Fly

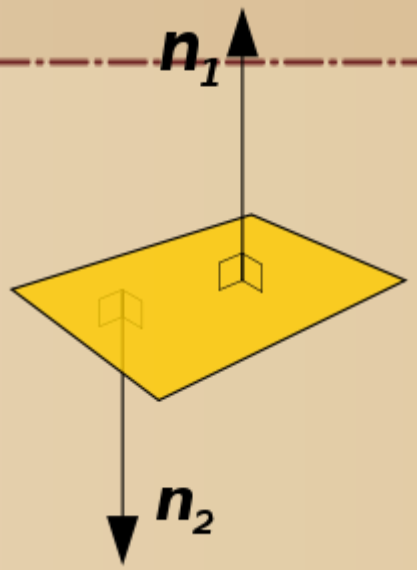
Analytical Geometry

Geometric notions such as distance and angle measure are defined using formulas.

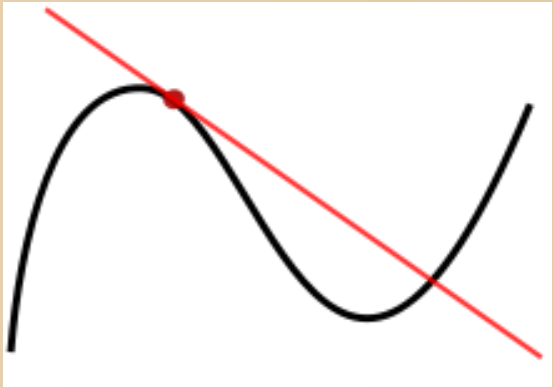
These definitions are designed to be consistent with the underlying Euclidean geometry.



Normal and Tangent Vectors



In geometry, a normal is an object such as a line or vector that is perpendicular to a given object. The normal is often used in computer graphics to determine a surface's orientation toward a light source for flat shading.



A straight line is said to be a tangent of a curve y = f(x) at a point x = c on the curve if the line passes through the point (c, f(c)) on the curve and has slope f'(c) where f' is the derivative of f.

Descartes Motion

Motion is the transference or transportation of bodies or parts of bodies in their relation to one another.

By tracking change in the coordinate, or Cartesian, system Motion can be expressed mathematically by plotting the parabolic curve of the projectile.

Descartes claimed to have deduced the mechanical structure of the universe from the mechanical principles of machines. - ***Deductive***

Orbital Move

Stability of an object sitting on the surface of the Earth is defined by whether movement causes the center of mass to move towards or away from center of Earth.

A body in a perfect circular orbit is continuously falling towards the attracting body but never comes closer to the attracting body.

Thus, if we slow down the satellite it falls to a lower orbit which becomes elliptical. If we accelerate the satellite it moves to a higher, elliptical orbit. Notice, when a space shuttle wants to return to Earth, it points its rockets in the direction of its orbit and fires them, thereby loosing speed so that it falls towards the Earth.

Bacon

Bacon became the English champion of inductive science –

- The purpose of acquiring knowledge is to improve human life; achieve happiness; and mitigate human suffering. In this sense, Bacon can be regarded as a humanist.
- In contrast to Descartes' deductive logic, Bacon emphasizes the inductive method, which ideally begins with observation and experimentation. This became the English Way. Ideally, hypotheses and theories are derived solely from observation and experimentation.

Deductive vs Inductive Science

Deductive Reasoning:

Start with a general statement, hypothesis or theory. Works its way down to a conclusion based on evidence.

Inductive Reasoning:

Start with observation or question. Work your way to a theory by examining related issues.